MODULE 10, LESSON 1

AI 101 for Mechanical Engineers

Protect the evaluation boundary

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Structural Outline

1. Foundational Architecture
2. Cross-Validation Mechanics
3. Boundary Selection & Leakage
4. Engineering Application Audit

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1. Foundational Architecture

Define constituent datasets and the objective of evaluation generalization.

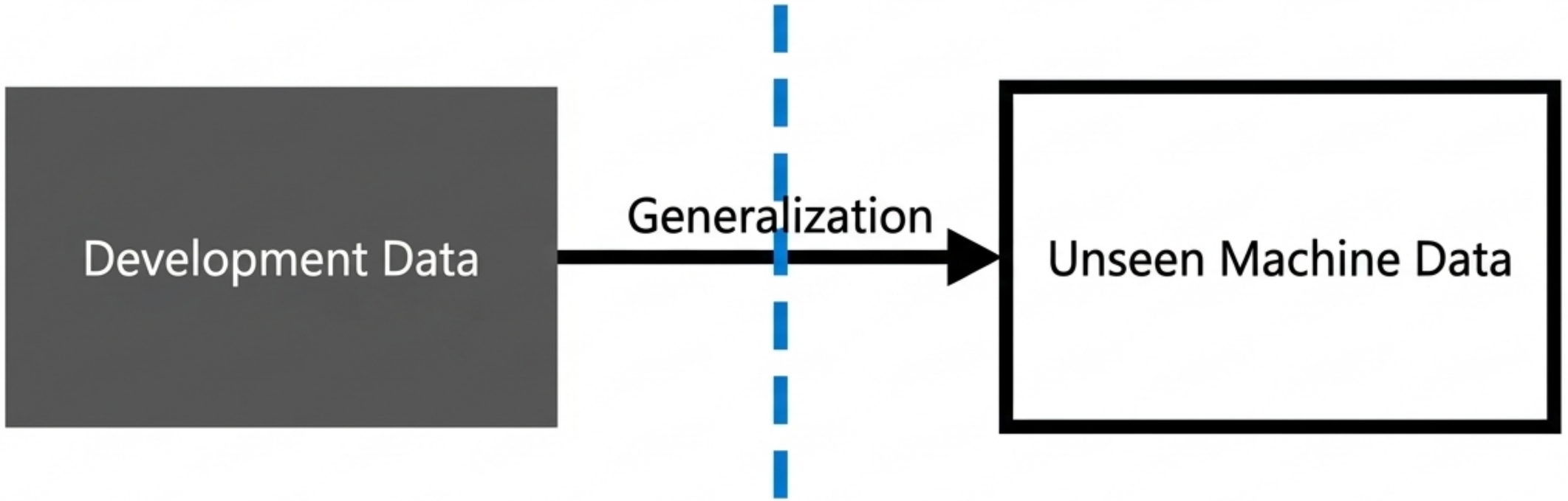
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Data Split Taxonomy

Training Set	Validation Set	Final Test Set
Purpose: Fit model parameters.	Purpose: Support choices like tree depth.	Purpose: Evaluate the locked model.
Boundary: Internal development data.	Boundary: Separate from parameter fitting.	Boundary: Strict prohibition on result tuning.

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Defining Generalization



- Generalization measures performance on relevant, completely unseen deployment examples.
- Testing evaluates if models function on a held-out machine.
- Repeatedly altering models using test results destroys this boundary.

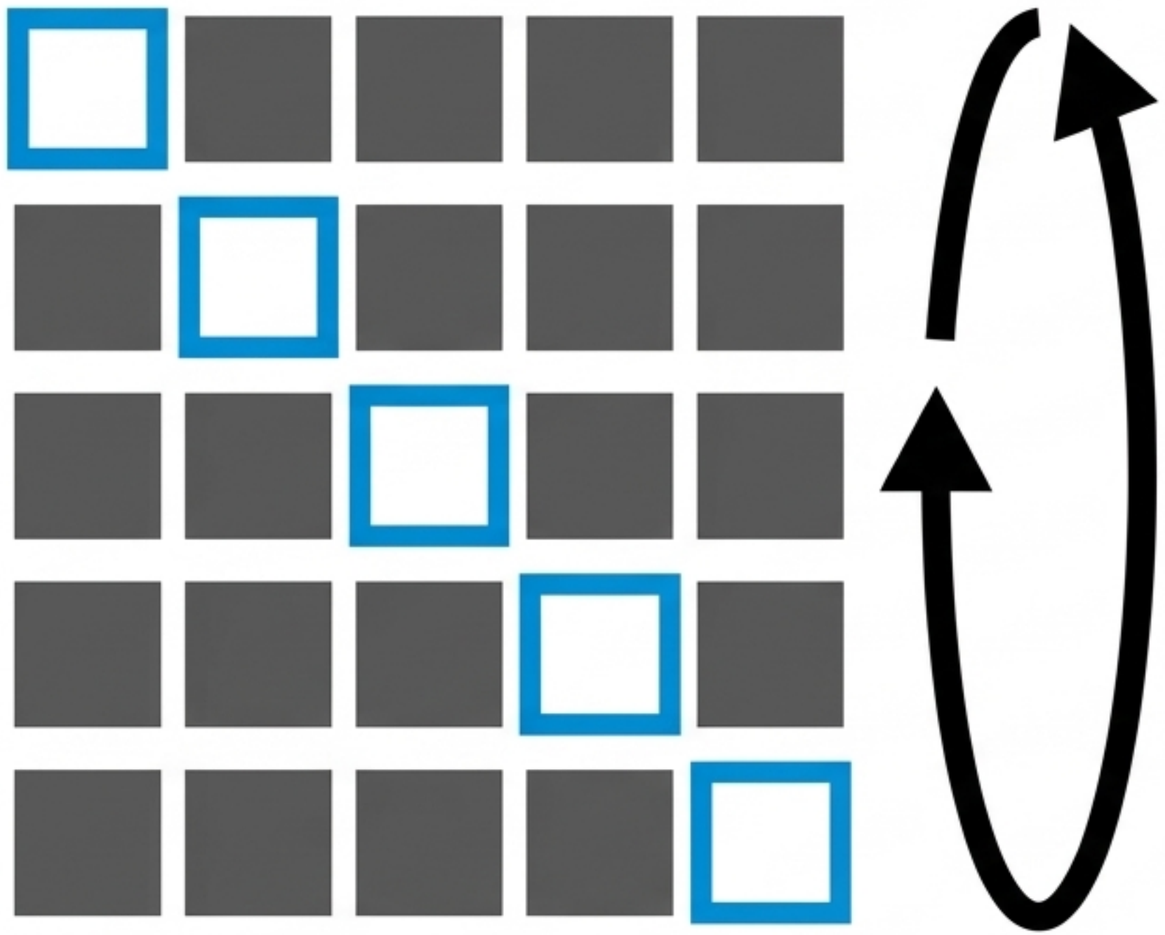
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2. Cross-Validation Mechanics

Detail the internal mechanisms of development dataset iteration.

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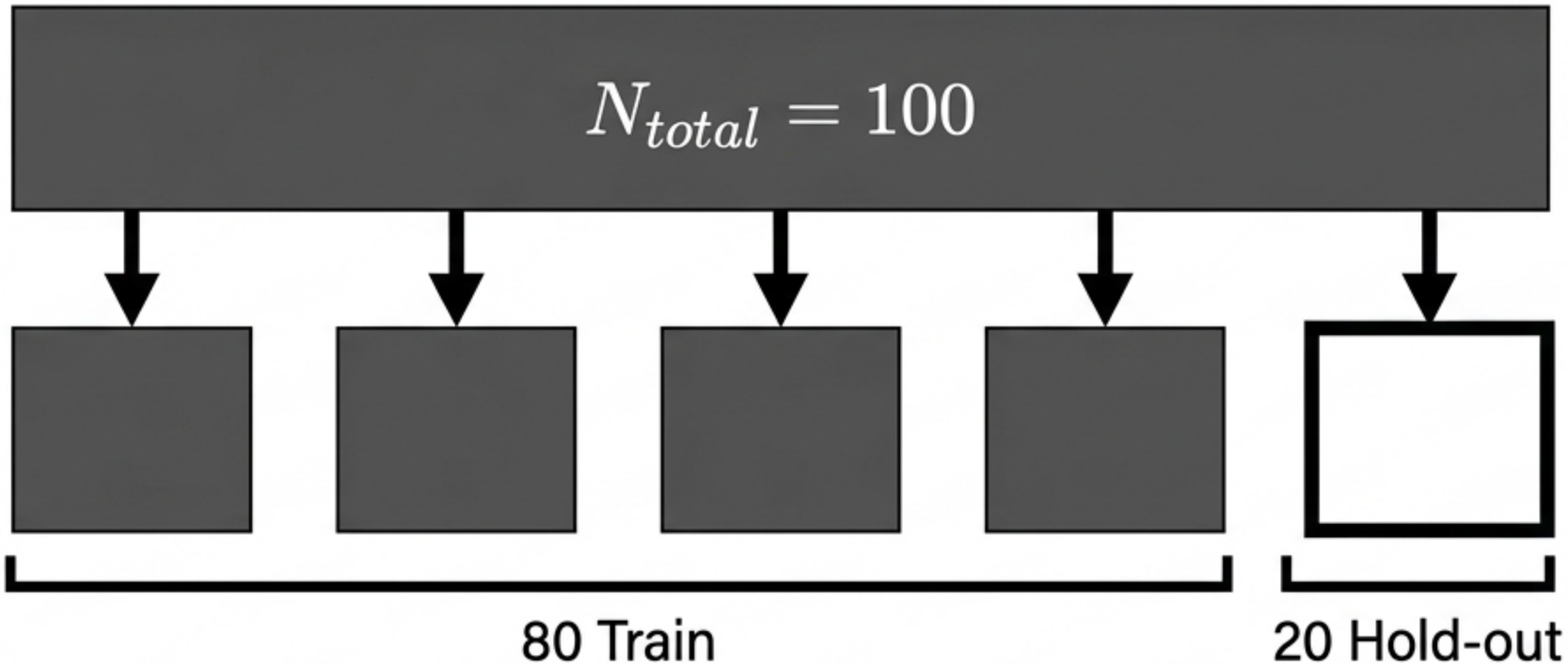
Internal Iteration Dynamics



- Cross-validation repeats splits exclusively within development data boundaries.
- Fit learned preprocessing separately inside each individual training fold.

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Cross-Validation Mathematics



- Assume five folds utilizing one hundred independent mathematical examples.
- Divide data equally into distinct, non-overlapping geometric fold arrays.
- Eighty examples populate the training set per fold iteration.
- Twenty examples occupy the validation hold-out per fold iteration.

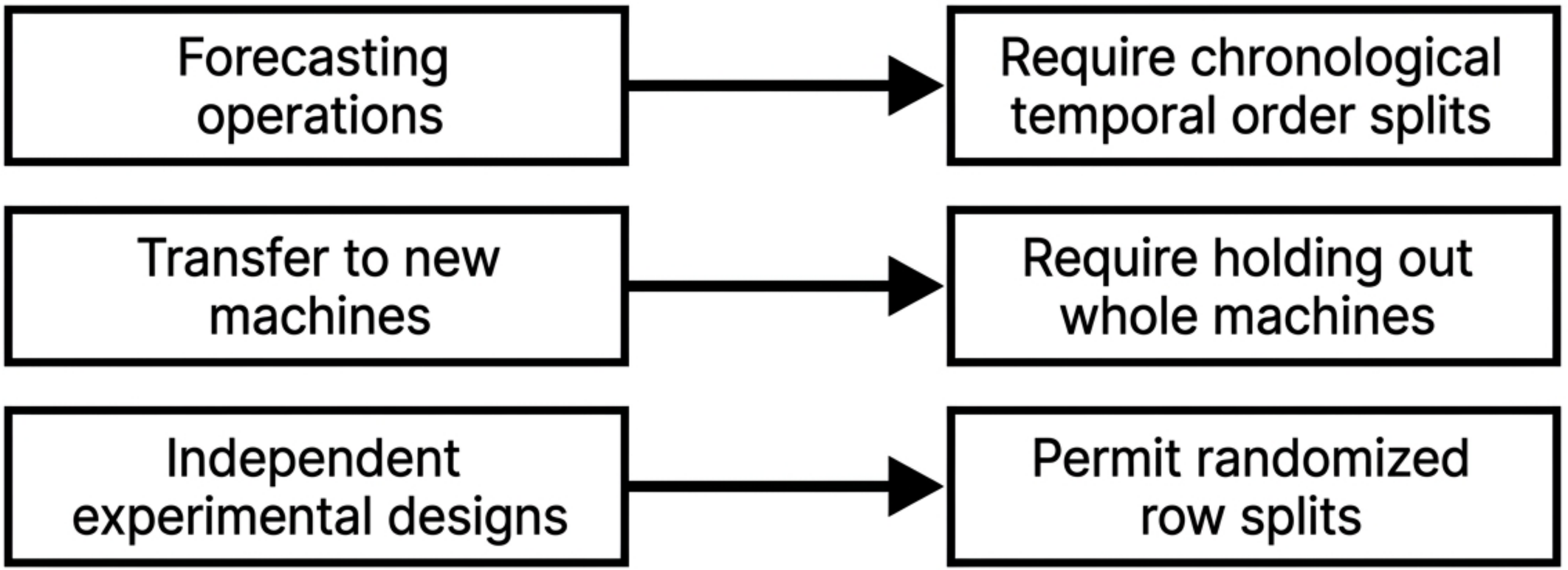
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3. Boundary Selection & Leakage

Establish rules for matching evaluation boundaries to specific engineering deployments.

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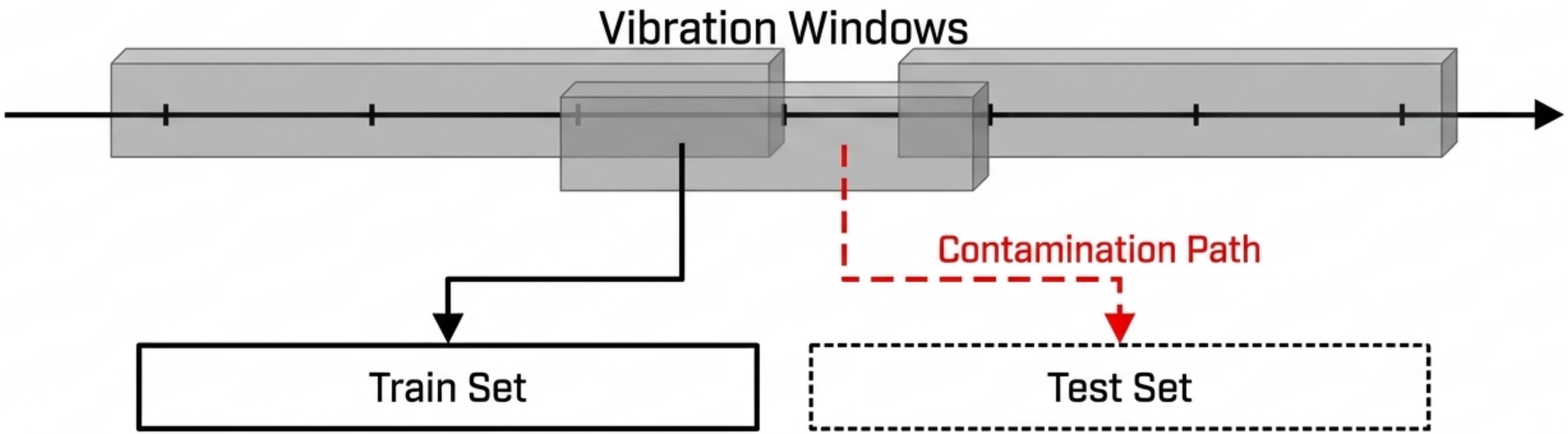
Deployment Mapping Matrix



- No single universal split rule exists for all datasets.
- Boundaries must exactly mirror the intended engineering deployment question.

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The Contamination Mechanism



- Overlapping vibration windows from identical events cause severe leakage.
- Presence in both sets yields artificially optimistic performance metrics.
- Row counts alone never guarantee mathematical or systemic independence.

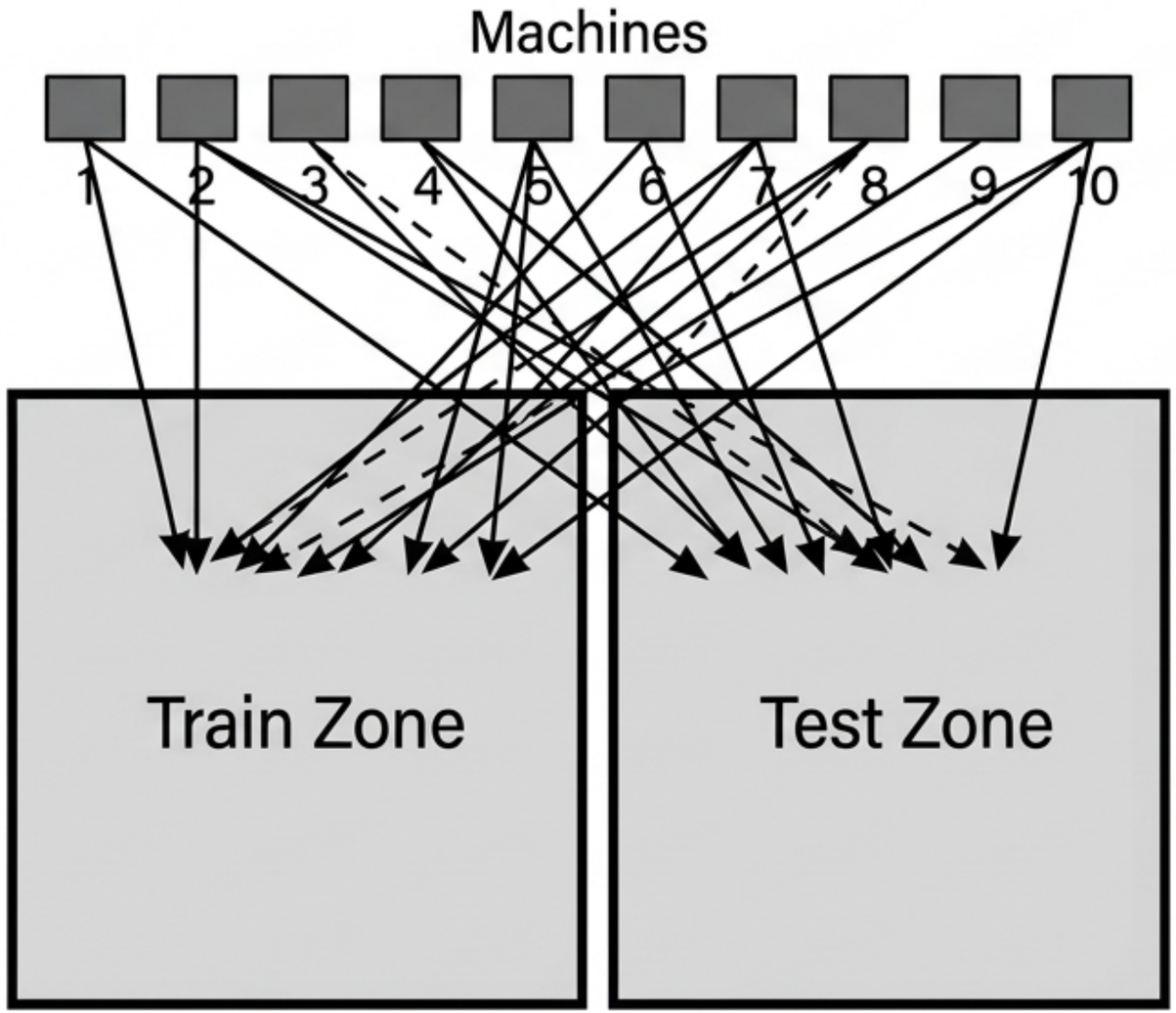
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4. Engineering Application Audit

Apply boundary axioms to audit a theoretical vibration dataset splitting strategy.

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State A: Faulty Boundary Setup



Dataset contains five thousand overlapping windows across ten machines.

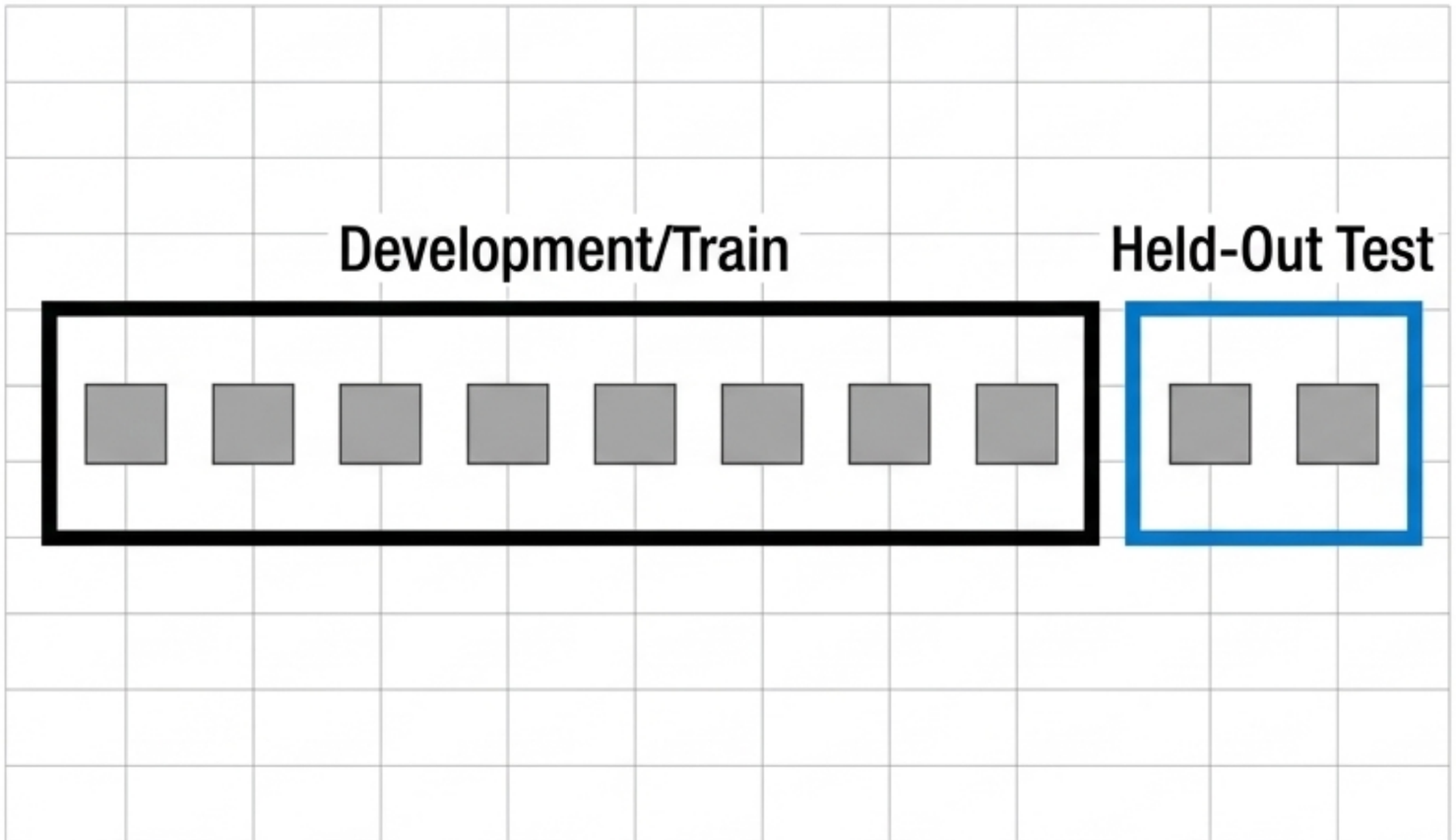
Deployment goal targets transfer to entirely unseen equipment.

Random row splits scatter every machine into both sets.

This architecture fundamentally **breaches the required independence boundary.**

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State B: Corrected Boundary Architecture



- Evaluate transfer capabilities by holding out entire whole machines.
- Group related overlapping events appropriately within consistent development boundaries.
- Fit learned preprocessing exclusively on the isolated training portion.
- Lock the selected model before evaluating the held-out equipment.

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Key Points

1. Test datasets mandate strictly locked models without further parameter tuning.
2. Cross-validation fits parameters and preprocessing exclusively within internal development folds.
3. Evaluation boundary architectures must precisely mirror the intended deployment question.
4. Evaluating equipment transfer necessitates strictly holding out entire, unseen machines.

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